

Software Maintenance Using LLMs

Objective

In this lab, you will perform software maintenance tasks using different Large Language Models (LLMs). You will analyze how effectively the models solve real maintenance problems such as bug fixing, feature enhancement, refactoring, and code improvement.

The source code of the project is given below:

Project Link:

<https://drive.google.com/file/d/1VJhVNEyoEkORD4cYrER1aQXK0qU64WwC>

Maintenance Tasks

Complete the following tasks using one or more LLM models.

No.	Task	Complaint/Request
1	SCROLL	Users complained that scroll bars don't always appear after painting outside the canvas, but when they do appear, the canvas doesn't look right.
2	YELLOW	Users complained that they can't select yellow.
3	UNDO	Users complained that the "Undo my last stroke" button doesn't always work.
4	LINE	Users requested a line tool. There's a radio button for it, but it doesn't work yet.
5	THICKNESS	Users requested control over the stroke thickness of the pencil, eraser, and line tools.

Requirements

For each task, you must:

- Use more than one LLM model (e.g., OpenAI ChatGPT, Google Gemini, Anthropic Claude, etc.)

- Write and refine prompts to solve the issue
- Apply the generated fix or improvement to the project
- Verify the fix by running/testing the project against the given complaint or request
- Compare the quality of solutions if multiple models are used

Report Submission

Your report must include the following for each task:

- Task name and complaint/request
- Prompt(s) used
- LLM model name and version
- Explanation of why the issue appeared (by LLM)
- Explanation of how the LLM solved the issue (by LLM)
- Where the fix was applied in the code (by LLM)
- Number of lines changed (by LLM)
- Testing/verification process and result
- Comparison between different model outputs
- Which solution you consider the best and why

You may include additional relevant observations if necessary.

Submit your report as a pdf file [BSSE14XX.pdf].